

ELECTRIC SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military Action Extension Methodology

DBA-ES: Electric Sector Parent Document

Governing Parent for DBA-ES-HY-FC1, DBA-ES-TX-FC2, DBA-ES-TX-FC3, DBA-ES-GC-FC4, DBA-ES-DI-FC5, DBA-ES-CntlCo, DBA-DC, DBA-ES-GN-FC6/7/8, and DBA-ES-POL

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For review by engineers, regulators, asset owners, and policy analysts responsible for protecting electric sector critical infrastructure.

Section 1 — Purpose and Scope

1.1 Purpose

This document establishes the Electric Sector Design Basis Framework. It is the sector-level parent instrument for a family of facility-class design bases that cover the principal high-consequence facility types within the electric sector: large hydroelectric storage facilities, extra-high-voltage transmission substations serving metropolitan load centers, high-voltage direct current interconnects, major natural gas compressor stations on interstate pipelines, and hyperscale data center campuses. Each facility class is addressed in a subordinate instrument that inherits the sector-level methodology established here and applies it to the specific physical systems, operational characteristics, and regulatory context of that class.

The framework is organized in three analytical layers: a baseline Design Basis (DB) that establishes postulated events derived from equipment failure, natural hazards, and operational conditions; a Design Basis Accident (DBA) layer that bounds the non-adversarial consequence envelope and establishes acceptance criteria; and a military action extension (DBA-MA) that applies the four distinguishing characteristics of deliberate adversarial action — intentionality, persistence, precision, and denial — to each postulated event category. This three-layer DB → DBA → DBA-MA progression is the logical architecture of the entire series. No facility-class instrument is complete without all three layers addressed, either within the instrument itself or by explicit cross-reference to a companion document.

The electric sector has operated without a formal design basis. Its mandatory reliability framework, the NERC Critical Infrastructure Protection (CIP) standards, addresses cybersecurity and, through CIP-014, physical security at a limited set of transmission facilities. It does not establish a design basis in the sense that nuclear safety engineering has employed it for six decades: a set of postulated conditions that the facility must be demonstrated to withstand, with defined acceptance criteria and a gap-analysis framework against which compliance can be measured. This document begins to supply that missing layer across the electric sector.

1.2 Scope

This document governs all facility-class design bases within the Electric Sector series. Scope is defined by the primary consequence chain: a facility belongs in the Electric Sector series if its highest-consequence failure mode primarily propagates through the electric grid or through services that depend on grid-delivered electricity. This organizing principle is examined in depth in Section 1.4.

The following facility classes are within the scope of this parent document:

- Large hydroelectric storage facilities, defined as facilities with installed generating capacity exceeding five hundred megawatts, whose primary storage reservoir presents a

downstream consequence potential in addition to the generating consequence of facility loss.

- Large extra-high voltage (EHV) transmission substations, defined as substations operating at three hundred forty-five kilovolts or above that serve metropolitan ring configurations, whose loss or damage would result in widespread instability, uncontrolled separation, or cascading within the interconnection under single- or double-contingency conditions.
- High-voltage direct current (HVDC) interconnects operating at capacities exceeding two thousand megawatts that provide cross-interconnection transfer capability or primary supply to large load centers.
- Major natural gas compressor stations on interstate transmission pipelines in corridors where station outage produces a measurable shortfall in gas supply to electric generation facilities whose combined capacity is material to bulk system reliability.
- Hyperscale data center facilities, defined as data center campuses exceeding one hundred megawatts of information technology load capacity serving multi-sector or government and military cloud workloads. The design basis for this class is established in DBA-DC (v1.6), which serves as the first facility-class application of this sector-level methodology.

The geographic scope is the United States bulk electric system and its principal dependencies. The methodology is transferable to allied-nation electric sector facilities without modification; jurisdictional annexes are anticipated but not yet drafted.

1.3 What This Document Is Not

This document is not a threat assessment. It does not characterize specific adversaries, assess the probability of any particular attack scenario, or provide intelligence regarding current threat actors. Those analyses are performed using the DBA-MA military action extension instruments, which derive their authority from this design basis.

This document is not an operational standard. It establishes the analytical envelope within which operational standards should be developed, not the operational procedures themselves. NERC CIP standards, TSA Pipeline Security Directives, FERC reliability orders, and facility-specific security plans are the instruments that translate design basis requirements into operational obligations.

This document is not a substitute for facility-class instruments. The sector-level analysis here establishes the methodology, the consequence taxonomy, and the inheritance model. Facility-specific postulated events, acceptance criteria, and gap analysis are addressed in the facility-class instruments listed in Section 8. No facility-class analysis should be performed directly against this document without the relevant facility-class instrument.

1.4 The Organizing Principle: Consequence Chain Over the NIPP Administrative Sector

The National Infrastructure Protection Plan organizes critical infrastructure into sectors primarily for administrative coordination: the sector structure answers which Sector Risk Management Agency coordinates federal partnerships with that sector's private owners and operators. That is a government organization question and an appropriate basis for interagency coordination. It is not an engineering question and is not an appropriate basis for organizing a design basis methodology.

Design basis methodology is consequence-chain-based. It asks: what conditions must this facility withstand, what happens if it fails to withstand them, and how far and through what pathways does the consequence propagate? Those questions are answered by physics and system topology, not by agency jurisdiction or sector administrative boundaries.

The mismatch between NIPP sector boundaries and engineering consequence chains is concrete in this series:

Large hydroelectric storage facilities are classified under the NIPP Dams sector, where the primary regulatory relationship is through the Army Corps of Engineers and the Bureau of Reclamation as dam safety authorities. The NIPP Energy sector governs the facility's generating function through FERC licensing and NERC reliability standards. For DBA-ES purposes, both consequence chains matter — the generating consequence (bulk system reliability) and the downstream consequence (dam failure flood wave) are both within scope. But the governing design basis analysis, the one that determines how the facility must be hardened and what conditions it must survive, runs through the electric grid. A large hydro facility is in this series because the loss of its generating function is a bulk electric system event.

Hyperscale data centers are classified under the NIPP Information Technology sector, where the regulatory conversation is primarily cybersecurity-centric, and the relevant agency is CISA. The DBA-ES series places data centers here because their governing consequence chain runs through the electric grid — their grid interaction characteristics (large block load, workload portability, cross-interconnection load transient potential), their physical security profile as large fixed facilities, and their foundational dependency relationship to all other critical infrastructure sectors are electric sector engineering questions. Their NIPP sector designation reflects their function; their DBA-ES placement reflects their consequence chain.

This organizing principle will require explicit framing in transmittal to agencies that consider a facility class within their sector. The message is not that NIPP sector designations are incorrect — they serve their coordination purpose. The message is that design basis methodology requires a different organizing principle, and that the DBA-ES series makes that principle explicit. Each facility class is addressed in this series because the primary consequence chain of its failure propagates through the electric grid. The sector risk management agencies for the NIPP Dams and NIPP Information Technology sectors remain the appropriate coordination points for their

respective regulatory relationships. This series provides the analytical standard that makes those regulatory relationships meaningful.

The empirical basis for this organizing principle at the defense-critical asset level is documented in the Defense Science Board Task Force on DoD Energy Strategy, “More Fight — Less Fuel” (February 2008). The DSB Task Force found that DoD identified 34 most critical assets — assets whose incapacitation or destruction would have a very serious, debilitating effect on the department’s ability to fulfill its missions — and that 31 of those 34 rely on commercial power grids as their primary source of electricity. The commercial power grid supplying those 31 critical assets is, in the majority of cases, DisCo infrastructure: distribution-level facilities that carry no NERC CIP-014 designation, do not trigger a grid cascade when they fail, and fall outside the consequence threshold that triggers TransCo-level reliability analysis. The DSB characterized those commercial grids as increasingly fragile and vulnerable. The DBA-ES consequence-chain organizing principle addresses this condition precisely: defense-critical consequence does not require grid-level cascade, and the analytical framework must be capable of reaching the DisCo assets that serve critical defense missions.

1.5 Applicability of the Military Action Extension

The military action extension developed in Section 5 and implemented in facility-class instruments is not universally applicable to all electric sector facilities. The DBA layer — Sections 3 and 4 of this document and the equivalent sections in each facility-class instrument — is the appropriate analytical terminus for the majority of CONUS electric sector facilities. The MA extension imposes additional analytical burden and hardening requirements that are warranted only where the threat environment, facility consequence profile, or infrastructure dependency characteristics justify them. Facilities that do not meet any of the trigger criteria defined below are DBA-complete; the military action extension is not required.

A facility is an MA extension candidate if it meets any one of the following six trigger criteria. Meeting a single criterion is sufficient to establish MA extension applicability; meeting none is sufficient to establish that the DBA layer is complete.

Trigger 1 — Consequence tier. The facility’s governing failure mode results in Tier 2 or Tier 3 consequences, as defined in Section 6. A facility whose worst-case failure is bounded within the facility boundary as a Tier 1 consequence — with no grid-level propagation and no cross-sector dependency impact — does not require MA extension. For most CONUS DisCo assets and smaller GenCo facilities, this criterion is the decisive filter. Their worst-case failure is a local outage, not a grid event, and the DBA is sufficient to handle it.

Trigger 2 — CIP-014 designation. Facilities carrying a NERC CIP-014 Tier 1 or Tier 2 designation are presumptive MA extension candidates. The CIP-014 consequence analysis, conducted by NERC and FERC, has already established that these facilities meet a threshold of grid consequence significance. That analysis is the predicate for the applicability of MA

extensions; it need not be repeated. Facilities with CIP-014 designations proceed directly to MA extension scoping.

Trigger 3 — Long-lead-time equipment. Facilities containing large power transformers at EHV voltage levels, or any equipment with a replacement timeline exceeding twelve months under normal procurement conditions, are MA extension candidates regardless of CIP-014 status. The persistence characteristic of military action — addressed in Section 5.2.2 — creates an extended-consequence window for long-lead-time equipment that the non-adversarial DBA does not bound. A TransCo substation containing a 500 kV autotransformer bank meets this trigger even if it does not satisfy the CIP-014 Tier 1 consequence threshold.

Trigger 4 — Military or government direct dependency. Facilities that directly supply electric power to military installations, continuity-of-government facilities, defense-critical manufacturing, or other infrastructure designated as defense-critical satisfy this trigger. Direct supply to a critical element in another sector establishes the applicability of MA extensions, whether or not the facility's loss triggers a grid cascade. This criterion reflects the GB approach to criticality — consequences for the dependent asset, not for the grid — and is the basis for the broadened Tier 3 definition in Section 6.1. The Defense Science Board documents the empirical scale of this trigger: of DoD's 34 most critical assets, 31 rely on commercial power grids — primarily DisCo distribution infrastructure — as their primary electricity source ("More Fight — Less Fuel," February 2008). Those 31 assets meet this trigger regardless of whether the DisCo facilities serving them would meet any other trigger criterion. The trigger fires on the consequence to the dependent asset, not on the consequence to the grid.

Trigger 5 — Cluster or corridor significance. Facilities within identified high-consequence clusters or critical transmission corridors are MA extension candidates because the shared infrastructure vulnerability characteristic means that individual facility consequence analysis at the DBA level understates the actual consequence. An adversary targeting the Northern Virginia data center supply infrastructure, a major metropolitan ring configuration, or an HVDC terminal station does not target individual facilities in isolation; it targets the shared infrastructure that serves the cluster. Facilities within such clusters or corridors meet this trigger based on their topological position rather than their individual consequence footprint.

Trigger 6 — Elevated geographic threat environment. Facilities in regions with documented elevated adversary presence, confirmed infrastructure pre-positioning activity, or geographic proximity to high-value military, nuclear, or government sites meet this trigger based on the threat environment rather than facility consequences alone. This criterion is the most judgment-dependent of the six and the most sensitive to intelligence context. For domestic CONUS facilities remote from military or national security infrastructure, this trigger will rarely fire. For allied-nation implementations — particularly in NATO member states on or near the eastern flank — it may be the primary trigger for a broad class of otherwise modest facilities.

The practical application of these trigger criteria is straightforward for most CONUS utility operators. A regional DisCo whose service territory does not include military installations or

defense-critical infrastructure, whose individual substations do not carry CIP-014 designations, and whose worst-case contingency is a local outage bounded at Tier 1, will not trigger any of the six criteria. The design-basis analysis for this series is complete at the DBA layer. The MA extension instruments do not apply to it, and no further analysis is required under this series.

GenCo operators of smaller hydro, wind, solar, or gas-fired generation facilities below the consequence threshold for CIP-014 designation, without direct military or government supply obligations, and outside identified high-consequence clusters, are similarly DBA-complete. The MA extension is warranted for large hydro storage facilities because of the downstream flood consequence and bulk generation significance — the triggers that define DBA-ES-HY-FC1 scope — not because all GenCo facilities require it.

CntlCo entities — ISOs, RTOs, and Balancing Authorities — present a distinct applicability question. Their physical infrastructure (control centers, communications systems, SCADA infrastructure) is not classified as a facility class in this series; DBA-ES-CntlCo (v0.2) is the functional class instrument for this series. It is a full three-layer instrument with DB, DBA, and MA extension layers complete. CntlCo infrastructure meets Trigger 5 (cluster/corridor significance) and, in most cases, Trigger 2 by analogy: the loss of CntlCo operational capability multiplies the consequence of simultaneous GenCo and TransCo attacks in ways the non-adversarial DBA does not address. The CntlCo functional class instrument is distinct from the physical facility-class design bases in this series. The SABD (Situational Awareness Blackout Duration) metric established in DBA-ES-CntlCo v0.2 §1.2 is the governing consequence metric for the CntlCo functional class. The MA extension — including the MA-1 state estimator corruption scenario and the AI governance attack surface convergence as a series first — is fully established in DBA-ES-CntlCo v0.2 §6 and §7.

Section 2 — Sector Profile

2.1 The Electric Sector as a Design Basis Domain

The electric sector presents a design-basis challenge with no direct precedent in sectors where formal design-basis methodology was developed. Nuclear safety engineering evolved its design-basis framework around a single facility type with consistent regulatory oversight, a well-defined physical-consequence domain (radiological release), and a clear hierarchy of barriers between the hazardous condition and the public. Electric sector facilities are heterogeneous in type, distributed across a continental-scale interconnected grid, and consequential primarily through the services they enable rather than through direct physical hazard.

This heterogeneity does not defeat the design basis concept — it requires adapting the concept's application at two levels. At the facility-class level, postulated events are specific to the facility's physical systems and operational characteristics. At the sector level, the design basis methodology establishes a common analytical framework: how postulated events are identified,

how consequences are taxonomized, what the acceptance-criteria structure looks like, and how gap analysis connects the analytical findings to capital investment and regulatory compliance.

The electric sector's defining characteristic for design-basis purposes is its foundational relationship with all other critical infrastructure sectors. Electric power is the substrate on which water treatment, petroleum refining, natural gas compression, financial clearing, emergency communications, military logistics, and digital infrastructure all depend. This substrate relationship means that the consequences of a failure at an electric-sector facility do not remain within the electric sector. They propagate outward to every dependent sector simultaneously. The consequence taxonomy throughout this series must explicitly account for this propagation.

Throughout this series, four terms identify the principal categories of electric sector entities by functional role. A GenCo (generation company) is any entity owning or operating generating facilities connected to the bulk electric system. A TransCo (transmission company) is any entity owning or operating high-voltage transmission facilities, including substations, switching stations, and transmission lines. A DisCo (distribution company) is any entity owning or operating lower-voltage distribution facilities that deliver power from the transmission system to end-use customers. A CntlCo (control company) is used in this series as the collective term for Independent System Operators (ISOs), Regional Transmission Organizations (RTOs), and Balancing Authorities — the entities responsible for real-time dispatch, frequency regulation, and transmission system reliability. In vertically integrated utilities, a single corporate entity may perform multiple functions; the terms refer to functional roles, not necessarily to separate legal entities. These terms are used throughout this document and all facility-class instruments in the series.

2.2 The Bulk Electric System Topology

The bulk electric system — the high-voltage transmission network that moves large blocks of power across the continental United States — is organized into three asynchronous interconnections: the Eastern Interconnection, the Western Electricity Coordinating Council (WECC), and the Electric Reliability Council of Texas (ERCOT). Each interconnection operates as a synchronously coupled system: frequency and voltage conditions propagate across the entire interconnection instantaneously. A significant disturbance anywhere in an interconnection is felt, to varying degrees, everywhere in that interconnection within seconds.

This synchronous coupling is the physical basis for the cross-facility consequence pathways that distinguish electric-sector design-basis analysis from facility-level analysis in isolation. The loss of a large GenCo asset, the failure of a critical TransCo substation, or the disconnection of a large block of load each produces a disturbance that propagates through the interconnection and requires a coordinated system response by the CntlCo. The design basis for any single facility must account for its role in that system response — not only for what happens to the facility, but for what the facility's loss or degradation does to the grid.

Three grid topology characteristics are governing conditions for design basis analysis across all facility classes in this series:

Shared infrastructure vulnerability is present across all facility classes. Transmission corridors, substations, and interconnection points serve multiple facilities simultaneously. Postulated events affecting shared infrastructure must be analyzed at the cluster or corridor level, not only at the individual facility level.

Asymmetric recovery timelines govern consequence severity. The bulk system can restore most contingency events within minutes to hours through automatic controls and operator response. Events that exceed the automatic recovery envelope — particularly those involving long-lead-time equipment such as large power transformers — produce consequences that persist on a timeline measured in months, not hours. The design basis for all facility classes must identify the conditions that cross from the rapid-recovery envelope into the extended-consequence regime.

Operational visibility gaps create consequence multiplication. CntlCos manage the bulk system through SCADA and energy management systems, which provide near-real-time visibility into grid conditions. Behind-the-customer-meter events — including the load behavior of hyperscale data centers, the protective relay operations of large industrial customers, and the automated responses of distributed resources — are invisible to the CntlCo until their effects appear on the monitored grid. Design basis analysis must account for the conditions under which visibility gaps amplify consequences.

2.3 Cross-Sector Dependency Profile

The electric sector's dependency relationships run in both directions. The sector depends on other infrastructure sectors to operate, and all other infrastructure sectors depend on the electric sector to function. Both directions of dependency are relevant to design basis analysis.

The electric sector's primary external dependency for operations is natural gas. Gas-fired generation currently provides the largest share of US electricity generation and provides virtually all of the flexible dispatchable generation that balances intermittent renewable supply. The gas-electric interdependency creates a bidirectional cascade risk: gas supply curtailment reduces the availability of electric generation, which in turn reduces power to electric-drive compressor stations and control systems that sustain gas transmission, further reducing gas supply. This cascade can self-sustain after the initiating event resolves if the secondary sector's degraded state prevents recovery in the primary sector. The gas-electric cascade is a governing cross-sector postulated event for the sector-level DBA.

The electric sector also depends on telecommunications infrastructure for grid control, protection system coordination, and CntlCo emergency communications. Loss of communications does not immediately cause a physical grid failure, but it degrades the CntlCo's ability to manage contingencies and execute restoration procedures. Telecommunications dependency becomes a

governing condition in scenarios where physical damage to GenCo or TransCo assets coincides with loss of monitoring and control.

In the outbound direction, the sector's dependency clients — water, petroleum, natural gas, financial, emergency services, military, and digital infrastructure — each have their own backup power postures, with durations ranging from minutes (grid-scale UPS systems) to hours (generator backup) to days (fuel-constrained generator operation). The design basis consequence taxonomy in Section 6 must map these backup postures to restoration timelines to establish the boundary between recoverable and catastrophic cross-sector consequences.

Section 3 — Design Basis: Electric Sector

3.1 The Design Basis Concept and Its Origins

The design basis is the set of postulated conditions that a facility must be demonstrated to withstand. It defines the envelope of credible events — the conditions the facility could plausibly encounter over its operational life — and establishes the safety system performance requirements and acceptance criteria that bound those events. The design basis does not assert that these events will occur; it asserts that if they occur, the facility must be capable of the specified response.

This concept was formalized in nuclear safety engineering over the decades following the Atomic Energy Act of 1954. The NRC's design basis regulations, embodied in 10 CFR Part 50 and the associated Standard Review Plans, require that nuclear facilities be designed and built to withstand a defined set of postulated accidents without loss of the ability to achieve and maintain a safe shutdown condition. The design basis analysis — which begins with the postulated events, traces the consequence chain through each safety system response, and demonstrates that the acceptance criteria are met — is the central analytical product of nuclear safety engineering.

No equivalent analytical framework has been applied to facilities in the electric sector. NERC reliability standards require that electric facilities be designed and operated to specified planning criteria — the N-1 and N-1-1 contingency standards that define the grid's required response to the loss of single and double elements. But planning criteria are not a design basis. They define the required system response to a loss event; they do not establish the envelope of postulated conditions that the facility must survive, the acceptance criteria for facility-level response, or the gap-analysis framework against which the current state can be measured. This document supplies the missing layer.

3.2 Three Required Adaptations

Transposing the design-basis concept from nuclear to electric-sector facilities requires three adaptations. These adaptations apply across all facility classes in the series. They are stated here at the sector level so that facility-class instruments can reference them without restating the underlying rationale.

First, the consequence taxonomy must extend beyond the facility boundary. Nuclear design basis accidents are analyzed primarily for their radiological consequences, which are contained by a defined hierarchy of physical barriers and evaluated against dose criteria at the site boundary. Electric sector facilities have no equivalent physical containment. Their consequences propagate outward through the electric grid, through dependent infrastructure systems, and through the downstream services those systems support. The design basis for an electric sector facility must formally account for consequences that extend beyond the facility fence line and establish criteria for the conditions under which facility-level consequences become grid-level or cross-sector consequences.

Second, the postulated event categories must reflect the specific physics and operational characteristics of the facility class. For a nuclear facility, postulated events are organized around the three critical safety functions: reactivity control, core cooling, and containment integrity. For electric sector facilities, the analogous organizing functions vary by facility class — power delivery continuity for a transmission substation, generation continuity for a hydroelectric plant, pipeline pressure integrity for a compressor station. Still, the analytical logic is the same: postulated events are those that challenge one or more critical functions in a manner that, if unmitigated, leads to a defined consequence category.

Third, the acceptance criteria must be expressed in terms that include operational states, not only physical parameters. Nuclear acceptance criteria are physical: peak cladding temperature, maximum reactor coolant pressure, and containment leak rate. Electric sector acceptance criteria must include operational state criteria: facility availability, service restoration timeline, grid contribution under contingency conditions, and cross-sector dependency satisfaction thresholds. These criteria have engineering content, as well as operational and economic dimensions that nuclear analysis does not address. The facility-class instruments must explicitly address these dimensions.

3.3 The Transferable Methodology Layer

Section 3 of each facility-class instrument in this series will implement this methodology for the specific facility class. The transferable layer — the elements that appear in substance across all facility-class instruments with facility-specific content substituted — consists of the following elements, which are established here at the sector level:

- Postulated event identification process: the structured process by which the universe of credible events is identified, organized into categories, and bound to the governing cases for each category.
- Consequence taxonomy structure: the three-tier taxonomy (facility-level, grid-level, cross-sector cascading) that organizes consequence analysis across all facility classes.
- Acceptance criteria framework: the structure for expressing acceptance criteria in physical, operational, and cross-sector terms at each consequence tier.

- Gap analysis framework: the methodology for comparing current facility capability against design basis requirements and expressing the gap as a structured finding that supports investment prioritization and regulatory compliance demonstration.
- Military action extension bridge: the analytical connection between the non-adversarial design basis and the DBA-MA extension, establishing the specific pathways by which the four military action characteristics modify each postulated event category.

Each facility-class instrument explicitly identifies which elements are inherited from this sector-level methodology and which elements are facility-class-specific. The boundary between inherited and locally defined content is a required section of every facility-class instrument.

3.4 Dual Design Basis Architecture

A facility within the Electric Sector series may simultaneously carry two distinct design bases that interact at the DBA level. The first is the sector-specific physical Design Basis established by this document and the applicable facility-class instrument: the set of postulated physical conditions — equipment failures, natural hazards, operational events, and cross-sector cascade scenarios — that the facility must be demonstrated to withstand. The second is an AI Governance Design Basis, established under the AI Governance series (Project 2), that governs the deployment of machine learning and generative AI systems in facility management, control, and operational support roles.

The two design bases are not independent. They interact at the DBA level: a postulated physical event may simultaneously stress the AI systems that support facility response, and an AI governance failure — loss of command-broker oversight, Gen AI non-deterministic behavior under stress, or ML model degradation outside its training envelope — may impair the facility’s ability to respond to a physical postulated event. Each facility-class instrument must acknowledge this dual-DB structure and identify the interaction points between the physical DBA and the AI Governance DBA that are specific to the facility class. The Bright Line and Command Broker architecture established in DBA-DC-L1 Section 3.5 and inherited by all facility-class instruments in this series is the primary mechanism for managing the boundary between the two design bases under normal and degraded operating conditions.

3.5 Sector-Level Postulated Event Categories

At the sector level, postulated events are organized into four categories that apply across all facility classes. Each category is instantiated in facility-class instruments with the specific events relevant to that class. The four categories and their governing rationale are:

Code	Category	Governing Rationale
PE	Power Supply and Grid Interface Events	All electric sector facilities interact with the bulk power system as a source, a load, or

		both. Conditions affecting the reliability, quality, or continuity of this interface are universal postulated event candidates.
SE	Structural and Physical Integrity Events	Fixed physical infrastructure is subject to equipment failures, natural hazards, and physical attacks. The consequence severity is governed by the facility's role in the grid topology rather than by the magnitude of the physical damage alone.
CE	Control and Communications Events	Electric sector facility operations depend on SCADA, protective relaying, and communications infrastructure. Disruption of control and communications creates operational blind spots that can propagate to physical consequences.
XC	Cross-Sector Cascade Events	The electric sector's foundational relationship with other critical infrastructure sectors means facility-level events can propagate across sector boundaries. Cross-sector cascade is a postulated event category at the sector level, even where it cannot be fully bounded at the facility level.

The XC category — cross-sector cascade events — is the one that most distinguishes electric-sector design-basis analysis from prior facility-specific frameworks. No NERC reliability standard, CIP requirement, or facility security plan has formally treated cross-sector cascade as a postulated event category requiring design basis treatment. This document establishes it as a required analytical element of every facility-class instrument in the series.

Section 4 — Design Basis Accidents: Non-Adversarial Envelope

4.1 Bounding Event Methodology

The non-adversarial design basis accident envelope consists of the postulated events that a facility must be designed and demonstrated to withstand, absent deliberate adversarial intent. These are the conditions caused by equipment failures, natural hazards, and operational errors — the events for which the traditional design-basis methodology was developed.

Bounding events are identified through a process that considers physical plausibility, historical frequency, and severity of consequences. Physical plausibility requires that the event be within the range of conditions that the facility's physical systems and location could produce. Historical frequency informs the credibility assessment but does not bound it: an event that has not occurred previously may still be within the design basis if engineering judgment supports its plausibility. Consequence severity determines which events are bounding cases. Within each postulated event category, the governing case is the event configuration that produces the highest consequence at each tier of the consequence taxonomy.

At the sector level, the following classes of bounding events apply across all facility classes. Facility-class instruments identify the specific events within each class applicable to the facility type and develop the bounding analysis for each.

4.2 Equipment Failure Events

Equipment failure events are the foundational category of the non-adversarial DBA. Every facility class has critical equipment whose failure challenges one or more critical safety functions. The design basis must identify the critical equipment, establish the failure mode that results in the highest consequence for each piece of equipment, and demonstrate that the facility can withstand that failure without exceeding the acceptance criteria.

Across facility classes, the governing equipment failure events share a common characteristic: they involve long-lead-time equipment whose replacement cannot be accomplished within a timeframe that would prevent escalating consequences. The large power transformer (LPT) at a TransCo EHV substation, the penstock and wicket-gate assembly at a GenCo large-hydro facility, and the high-pressure compressor unit at a critical compressor station each represent this long-lead-time equipment class. The design basis must treat these as special categories requiring dedicated analysis of the extended-consequence regime that their failure can produce.

4.3 Natural Hazard Events

Natural hazard events are postulated based on the facility's geographic location and the historical record of the hazard at that location. The governing natural hazards across electric sector facility classes are seismic events, extreme wind events (including hurricanes, tornadoes, and derechos),

flooding (riverine and storm surge), extreme temperature events (polar-vortex cold, heat waves), and wildfire-smoke events affecting transmission line ratings and substation equipment.

The natural hazard design basis must address compound events — conditions in which two or more hazards occur in proximity in time and in which recovery from the first hazard is impeded by the second. Winter Storm Uri (February 2021) illustrated the compound-event problem at the sector scale: cold weather stress coincided with fuel-supply disruptions for gas-fired generators, and recovery from generator trips was impeded by the same weather conditions that caused them. The natural hazard DBA must include at least one governing compound event scenario for each facility class.

4.4 Operational Events

Operational events are postulated conditions arising from human error, procedural deviation, or control system malfunction in the absence of physical equipment failure or external hazard. The governing operational events across electric sector facility classes include protective relay misoperation, control system failure or erroneous actuation, human error during maintenance that results in equipment damage, and software failure in energy management or SCADA systems.

Operational events require particular attention in the DBA-MA extension because they can be induced deliberately: a sophisticated adversary who understands a facility's control systems can engineer an operational event that mimics an accidental condition while achieving a targeted physical consequence. The non-adversarial DBA establishes the envelope for accidental operational events; the DBA-MA extension applies the intentionality characteristic to that same envelope.

4.5 Cross-Sector Cascade Events

Cross-sector cascade events are the sector-level category most distinct from prior facility-class design basis practice. A cross-sector cascade event is a condition where disruption of one sector's infrastructure produces a consequence that impairs the electric sector's ability to operate, which in turn impairs another sector's ability to operate, creating a bidirectional cascade that neither sector can resolve independently.

The gas-electric cascade is the governing case. Interstate natural gas pipelines depend on electric power for compression; electric generation depends on natural gas for fuel. A cold weather event that simultaneously increases heating demand (and therefore gas consumption) and reduces gas compression capability (due to power availability constraints for electric-drive compressors) can produce a cascade in which a gas supply shortfall reduces generation availability, which in turn reduces the power available for compressors, further reducing gas supply. This cascade is not bounded by either the gas sector's or the electric sector's design basis when analyzed in isolation. It requires explicit cross-sector DBA treatment.

Facility-class instruments must identify the specific cross-sector cascade pathways relevant to the facility class and develop the governing case analysis for each pathway. The governing case is the cascade configuration that produces the largest cross-sector consequence within the DBA-MA planning horizon.

Section 5 — The Military Action Extension

5.1 Why the Non-Adversarial Design Basis Is Insufficient

The non-adversarial design basis defined in Section 4 is necessary but insufficient as a complete analytical framework for electric sector facilities. Each bounding event in Section 4 shares a characteristic that distinguishes it from the military action threat tier: it is not chosen. Equipment fails according to probabilistic distributions. Polar vortices follow meteorological physics. Seismic events occur along fault lines. The facility designer can analyze historical frequencies, model consequence chains, and invest in protection against the events most likely to occur within a defined planning horizon. The adversary does not operate within this framework.

Military action against electric-sector infrastructure introduces four characteristics absent from the non-adversarial design basis that require an extended analytical framework to address. These four characteristics were identified in the DBA-MA parent framework in the nuclear context and subsequently confirmed in applications across the natural gas, water, petroleum, and data center sectors. They apply to the electric sector without modification. Their application to each facility class in this series is the work of the DBA-MA military action extension instruments.

5.2 The Four Military Action Characteristics

5.2.1 Intentionality

The adversary chooses its targets. It does not attack at random or according to a probabilistic distribution. It studies the grid topology, identifies the highest-consequence nodes, assesses the defensive posture of candidate targets, and selects the attack configuration most likely to achieve the desired strategic effect. For the electric sector, intentionality means the adversary will not attack a hardened TransCo substation when an equivalent node with weaker physical security achieves the same consequence. It means the adversary will identify the shared infrastructure that serves multiple facilities simultaneously — the transmission corridor, the water supply, the communications backhaul — and target it rather than the individual facility. Intentionality also means the adversary will consider CntlCo infrastructure — the energy management systems, SCADA communications, and dispatch centers that coordinate grid response — as targets whose disruption would multiply the consequences of simultaneous physical attacks on GenCo and TransCo assets. Intentionality transforms the probability-based frequency analysis of the non-adversarial DBA into a capability-based worst-case analysis for the DBA-MA extension.

5.2.2 Persistence

Military action does not resolve on its own timeline. Equipment failure is a point event; the consequence cascade begins to resolve as soon as the failed equipment is replaced or bypassed. Military action can persist indefinitely: the adversary can continue to deny access for repairs, target replacement equipment before installation, and sustain the consequence-producing condition beyond the timeline the facility's design basis for spontaneous failure was intended to address. For long-lead-time equipment — large power transformers, high-capacity generator turbines, HVDC converter hardware — persistence combined with replacement lead times of twelve to twenty-four months creates a consequence window that has no analog in the non-adversarial DBA.

5.2.3 Precision

Modern state-level military action can target specific systems with sufficient precision to achieve targeted functional degradation rather than simple physical destruction. An adversary who understands a TransCo substation's protection coordination architecture can target the relay systems that would isolate a fault, leaving the facility physically intact but unable to protect itself under stress conditions. An adversary who understands a GenCo hydroelectric facility's control systems can target the governor and excitation control in ways that are analytically indistinguishable from a software fault until the consequence materializes. Precision means that the adversary is not bound by the collateral damage considerations that constrain indiscriminate weapons; it can achieve the maximum consequence with the minimum footprint.

5.2.4 Denial

Military action can deny access to the site for assessment, repair, and emergency response. The denial characteristic is the one that most fundamentally breaks the recovery assumptions embedded in the non-adversarial DBA. Every non-adversarial design-basis analysis assumes that qualified personnel can reach the facility within a defined time window and that the replacement parts supply chain is available. Under conditions of military action, neither assumption may hold. Access denial transforms a manageable equipment failure into an extended-consequence event. For any facility where the governing consequence depends on the restoration timeline, denial is the characteristic that determines whether the consequence remains bounded or escalates.

5.3 The Extended Design Basis: DBA-MA

The DBA-MA extension takes each postulated event category in the non-adversarial DBA. It applies the four characteristics of military action to identify the governing military action scenario for each category. The extension is not a separate event list — it is an analytical overlay on the non-adversarial DBA that asks, for each postulated event: what does this event look like when an adversary chooses the initiating condition, sustained for the adversary's timeline,

targeted at the highest-consequence configuration, and accompanied by denial of the recovery actions the facility's design relies upon?

For each postulated event category, the DBA-MA extension produces a governing military action scenario that represents the most severe credible adversarial exploitation of that category. The governing scenario is the one against which the facility's hardening measures and response procedures must be demonstrated to be adequate. The facility-class instruments develop the governing military action scenarios for each category within its specific facility-class context.

The DBA-MA extension is addressed at the sector level here and in the facility-class instruments through the bridge section that connects the non-adversarial DBA to the military action analysis. The sector-level military action analysis — addressing multi-facility coordinated attacks, cross-interconnection consequence, and supply chain interdiction at the national scale — is developed in the sector-level DBA-MA companion to this document, which is anticipated as the next series of instruments following completion of the first facility-class design bases.

Section 6 — Consequence Taxonomy

6.1 Three-Tier Structure

Consequences across all facility classes in this series are organized into a three-tier taxonomy that reflects the pathways by which facility-level events propagate to grid-level and then cross-sector events. The three tiers are not sequential stages — a severe event can produce Tier 3 consequences without fully traversing Tier 1 and Tier 2. They are analytical categories that correspond to the geographic and sectoral scope of the consequences.

Tier 1: Facility-level consequences. These are consequences bounded within the physical perimeter of the facility and its immediately connected transmission or pipeline infrastructure. Equipment damage, loss of generating capacity, and loss of transmission capacity that do not trigger stability concerns in the wider interconnection are Tier 1 consequences. The primary design basis — the non-adversarial DBA — establishes acceptance criteria at the Tier 1 level: the facility must be capable of containing Tier 1 consequences within the facility boundary.

Tier 2: Grid-level consequences. These are consequences that cross the facility boundary and affect the reliability, stability, or capacity of the bulk electric system within the interconnection. Frequency excursions, voltage collapse in a load area, loss of a major TransCo transmission corridor, resulting in thermal overloads on adjacent paths, and the triggering of under-frequency load shedding are Tier 2 consequences. Tier 2 consequences require coordinated system response by the CntlCo, the TransCo, and other resources across the interconnection. The acceptance criteria at Tier 2 establish the conditions the facility must not produce in the interconnection: it must not, by its failure mode alone, trigger Tier 2 consequences that the grid is not designed to absorb.

Tier 3: Cross-sector consequences. These are consequences that propagate beyond the electric grid into dependent infrastructure sectors. Two pathways reach Tier 3, and either is sufficient to establish Tier 3 applicability for a given facility.

The first pathway is a cascade: consequences that propagate through grid-level degradation into dependent sectors. Water treatment plants going to backup power, hospital generators activating, natural gas compression degrading as electric supply fails, financial clearing systems transitioning to backup data centers, and military logistics systems losing primary computing infrastructure are all Tier 3 cascade consequences. This is the pathway captured by the NERC CIP-014 consequence logic, which qualifies facilities based on their potential to produce widespread instability or cascading within the interconnection.

The second pathway is direct critical dependency: consequences that reach a critical function in another sector through direct supply disruption, whether or not a grid-level cascade occurs. A TransCo substation whose loss directly removes power from a military installation, a water treatment facility, or a continuity-of-government site produces a Tier 3 consequence through direct supply interruption, even if the substation's protection system isolates the fault cleanly and no cascading event occurs in the interconnection. This pathway reflects the approach taken in Great Britain's critical infrastructure security framework, which designates an asset as Critical if it supplies power to a critical element in another sector, irrespective of cascade potential. The domestic US evidentiary basis is established by the Defense Science Board Task Force on DoD Energy Strategy ("More Fight — Less Fuel," February 2008), which documented that 31 of DoD's 34 most critical assets rely on commercial distribution grid infrastructure — DisCo assets carrying no cascade-based regulatory designation — as their primary electricity source, and characterized those grids as increasingly fragile and vulnerable. The DBA-ES series adopts the direct dependency standard alongside the cascade pathway: both routes to Tier 3 are analytically equivalent for purposes of consequence taxonomy and MA extension applicability.

Tier 3 acceptance criteria establish the maximum tolerable duration of electric service disruption for each dependent sector class before that sector's backup posture is exhausted and unrecoverable consequences begin. This duration-based criterion applies to both Tier 3 pathways. The facility-class instruments must identify which Tier 3 pathway governs for each facility class and develop the acceptance criteria accordingly.

6.2 Long-Lead-Time Equipment and the Extended Consequence Regime

The standard consequence taxonomy assumes that Tier 1 consequences are bounded and that recovery from Tier 2 and Tier 3 consequences proceeds within hours to days as the grid operator activates contingency resources and the facility begins restoration. This assumption holds for most equipment failures and for natural hazard events where the hazard itself is time-bounded.

Long-lead-time equipment failure — with or without adversarial action — breaks this assumption. Large power transformers at TransCo EHV substations are the governing case. LPT lead times for new custom units currently range from eighteen to thirty months. Spare

transformer programs (the DoE’s NERC Spare Transformer Equipment Program — STEP) provide a limited inventory of pre-positioned spares, but STEP coverage does not address all critical EHV transformer configurations. A TransCo substation loss driven by LPT damage produces a consequence that, absent a compatible spare available for rapid installation, persists on a timeline incompatible with the grid’s designed contingency recovery.

This extended consequence regime is a specific analytical category required in facility-class instruments for EHV substations and for any other facility class where governing equipment has comparable lead times. The acceptance criteria for the extended consequence regime must include not only the facility-level restoration timeline but also the grid reliability posture during the extended outage period: which compensating measures are required, which transmission constraints must be managed, and which cross-sector dependency thresholds are at risk within the restoration window.

The recovery timeline for the governing long-lead-time component is designated **RT_HVLLC** (Recovery Timeline for High-Value Long-Lead Components) in this series. RT_HVLLC is defined as the time required to restore a facility to service following destruction of its most difficult-to-replace major component, expressed in months. For transmission-connected generation facilities and TransCo EHV substations, the binding constraint is the high-voltage main power transformer: a custom-manufactured component with typical lead times of twelve to twenty-four months under normal commercial conditions and no meaningful strategic reserve in North America. RT_HVLLC represents the maximum duration of forced outage following a precision physical attack on the facility’s most vulnerable single component and is the primary consequence-duration metric in the DBA-MA extension for any facility class containing long-lead-time equipment. This construct originated in the GenMix Optimization Whitepaper (Roxey, Eclectic Technologies, 2026) and is adopted here as a series of standard metrics.

6.3 Cross-Sector Consequence Mapping

The Tier 3 consequence analysis requires a mapping of each facility class to its primary cross-sector dependency clients. This mapping is established at the sector level here and refined in facility-class instruments based on the specific geographic and grid-topology context of the facility class.

The cross-sector dependency clients and their governing backup posture durations, at the national planning level, are as follows. These figures represent the general posture of each sector class and will vary by specific facility; facility-class instruments must develop a site-specific version of this analysis for the facility class’s geographic service territory.

Hospitals and medical facilities: backup generator capacity typically provides seventy-two to ninety-six hours of primary operations. Fuel delivery dependency becomes governing after the first twelve hours as fuel reserves are consumed.

Water treatment and distribution: electric-drive pumps at water treatment plants typically have backup generator capacity of 24 to 72 hours, depending on the facility's investment posture. Loss of water pressure for fire suppression compounds consequences, independent of the water treatment function.

Natural gas transmission compression: as analyzed in the DBA-GAS series, electric-drive compressor stations have limited backup capacity. Stations on critical corridors may lose compression capability within hours of sustained grid loss, triggering the gas-electric cascade described in Section 4.5.

Financial sector: major financial infrastructure has invested heavily in backup power and geographic redundancy. The governing vulnerability is not the primary backup duration but the period beyond which geographic redundancy is exhausted, affecting both primary and backup facilities.

Military and government operations: federal facilities have diverse backup postures. The governing DBA-MA concern is not the routine backup duration but the conditions under which adversary-induced electric-sector disruption is coordinated with other operations to degrade military response capability.

Section 7 — Protective Action Principles

7.1 Layered Protection Architecture

The protective action architecture for electric sector facilities under both non-adversarial and military action conditions follows a defense-in-depth logic: no single protective measure is sufficient, and measures are layered such that the failure of any individual layer does not compromise the facility's ability to meet its acceptance criteria. This logic, developed in nuclear safety engineering and confirmed in the DBA-MA nuclear series, applies without modification to facilities in the electric sector.

The layers are sequenced from passive physical hardening (the baseline, which requires no action to be effective) through active operational response (procedures and personnel) to external coordination (grid operator, government, military). The local sovereignty principle, established in the DBA-MA nuclear series, also applies to electric sector facilities: the facility's protective measures must be capable of sustaining critical functions without external power, communications, or operator action for the design-basis duration. Active and external response layers add resilience but cannot substitute for passive physical hardening as the foundation.

7.2 Command Broker and Bright Line Requirements

Automated control systems increasingly govern the protective responses of facilities in the electric sector. SCADA-driven load shedding, automatic protective relay operation, and software-controlled governor response all involve automated actions on physical infrastructure

that can have consequences if the automation operates incorrectly or is manipulated by an adversary.

The Bright Line, as defined in FD-BL (Foundational Definition — The Bright Line), applies to the electric sector. Under FD-BL §4.2–§4.3 the Bright Line is a threshold on individual actions: an action taken on a physical process parameter is above the line when its incorrect performance could produce a consequence at or above the facility's severity threshold, and below the line when it could not. Placement follows the consequence of the action, not the technology that produces it — a deterministic control action can be above the line, and a generative recommendation can be below it. Determinism functions as the scope carve-out of FD-BL §4.4, not as the placement test: a component that is deterministic within a stated input bound and formally testable as deployed is outside the Command Broker obligation, whatever its role, while a generative or otherwise non-deterministic component is within scope and its above-line actions require a human Command Broker enforced architecturally (FD-BL §4.5). A human command broker is required for every above-line physical process parameter change commanded by or through a generative AI system.

This requirement applies to all facility classes in the DBA-ES series. Facility-class instruments must identify the specific physical process parameters that exceed the Bright Line for their facility class and define the command-broker architecture required to satisfy the Bright Line requirement.

7.3 Recovery Sequencing

Recovery from design basis events — both non-adversarial and military action — proceeds through a defined sequence that facility-class instruments must establish for each class. At the sector level, the governing recovery sequencing principle is grid stability first: no recovery action at the GenCo or TransCo facility level proceeds in a manner that introduces instability into the interconnection. The CntlCo's coordinated recovery plan governs the sequence and timing of facility restoration; individual GenCo and TransCo operators execute it.

Under military action conditions, recovery sequencing is further constrained by the characteristic of access denial. Facility-class instruments must identify the recovery actions that require on-site presence and establish the minimum access requirement for each. The DBA-MA extension for each class must then analyze the scenario in which access is denied for the design-basis denial duration and demonstrate that the facility can remain in a bounded-consequence state for that duration without recovery action.

Section 8 — Facility Class Register and Inheritance Model

8.1 The Inheritance Model

Each facility-class instrument in the Electric Sector series inherits the sector-level methodology established in this document and applies it to the specific physical systems, operational characteristics, and regulatory context of the facility class. The inheritance relationship is explicit and structural: facility-class instruments do not re-derive the methodology; they implement it.

The inheritance model operates as follows. The sector-level document establishes: the organizing principle (consequence chain over NIPP sector); the DB → DBA → DBA-MA analytical progression; the three required adaptations for electric sector application; the four sector-level postulated event categories; the three-tier consequence taxonomy; the acceptance criteria framework structure; the gap analysis framework structure; the Bright Line and command broker requirements; and the recovery sequencing governing principles. These elements appear in substance in every facility-class instrument, with facility-specific content substituted where the sector-level formulation requires it.

Each facility-class instrument must include a section explicitly identifying which elements are inherited from this document unchanged, which elements are inherited with facility-class-specific adaptation, and which elements are entirely locally defined within the facility-class instrument. This inheritance map is a required component of every facility-class instrument at the time of first publication. It must be updated at each revision to reflect any changes to the sector-level document.

8.2 Facility Class Placement Rationale

The following identifies the specific rationale for each facility class's placement in the DBA-ES series under the consequence-chain organizing principle. Placement rationale is stated here so that facility-class instruments can cross-reference it without restating the argument.

Hyperscale data centers (DBA-DC) are placed in the DBA-ES series because their primary consequence chain runs through the electric grid. Their grid interaction characteristics — large-block load, workload portability producing simultaneous load dropout and surge, and cross-interconnection load-transient potential — are questions in electric system engineering. Their NIPP sector designation (Information Technology) reflects their function; their DBA-ES placement reflects their consequence chain. The data center sector's complete absence of mandatory physical security standards, and the regulatory gap that absence represents, is an electric-sector consequence question because the facilities are electric-sector load resources at the scale of medium-sized generating units.

Large hydroelectric storage facilities (DBA-HY) are included in the DBA-ES series because their primary generating consequence—the loss of bulk synchronous GenCo generation—is a bulk electric system reliability event. The NIPP Dams sector governs the dam-safety relationship (dam failure, flood wave, reservoir integrity), while DBA-HY addresses the chain of consequences that propagates through the electric grid. The two consequence chains are analyzed together within DBA-HY because they share the same initiating events under military action conditions: the weapon that damages the penstock causes both a loss of GenCo generation and a potential dam safety consequence.

Large EHV transmission substations (DBA-ES-TX-FC2) are placed in the DBA-ES series as a native TransCo facility class. They have no NIPP-sector ambiguity — they are Energy-sector facilities. Their placement in the series reflects their governing vulnerability: TransCo EHV

substations at metropolitan ring configurations are the single facility class for which long-lead-time equipment exposure, combined with the intentionality and precision characteristics of military action, produces the most severe extended-consequence scenario in the electric sector.

HVDC interconnects (DBA-ES-TX-FC3) are placed in the DBA-ES series as native TransCo infrastructure. Their governing design-basis characteristic is their cross-interconnection role: the loss of a major HVDC interconnection removes the transfer capability that allows one interconnection to support another under stress. Under military action conditions, targeting TransCo HVDC infrastructure is a strategy for isolating a geographic region from grid support. This strategic consequence has no analog in the non-adversarial DBA.

Major natural gas compressor stations (DBA-ES-GC-FC4) are placed in the DBA-ES series because their governing consequence chain runs through the gas-electric cascade. The DBA-GAS series addresses natural gas infrastructure comprehensively from the perspective of the gas sector's own design basis. DBA-ES-GC-FC4 addresses the subset of compressor stations whose loss results in a measurable consequence in the electric sector through gas-supply disruption to GenCo gas-fired generation. The two analyses are complementary: DBA-GAS establishes the gas sector design basis, and DBA-ES-GC-FC4 establishes the electric-sector consequence analysis for the compressor stations in critical GenCo generation-support corridors.

8.3 Facility Class Register

The following table is the current facility class register for the Electric Sector Design Basis Series.

Subsector	Designation	Facility Class	NIPP Sector	Status	Primary Regulatory Anchor
Critical Loads (DBA-ES-CL)	DBA-DC	Hyperscale data center (>100 MW IT load)	Information Technology	Active — v1.7	Uptime Tier IV; NERC CIP customer meter boundary
—	DBA-ES-HY-FC1	Large hydroelectric — storage (>500 MW)	Dams / Energy	In development	FERC Part 12; NERC CIP-014; USCOLD safety guidelines
—	DBA-ES-TX-FC2	Large EHV transmission substation (345 kV–500 kV, metropolitan ring)	Energy	Anticipated	NERC CIP-014 Tier 1; FERC Order 693 TPL standards

		configuration)			
—	DBA-ES-TX-FC3	HVDC interconnect (>2,000 MW, cross-interconnection)	Energy	Anticipated	FERC; NERC FAC-002; applicable TPL reliability standards
—	DBA-ES-GC-FC4	Major natural gas compressor station (interstate pipeline, critical corridor)	Energy	Anticipated	TSA Pipeline Security Directives; FERC; PHMSA
—	DBA-ES-DI-FC5	Distribution substation with edge generation (HV/EHV input; 69–110 kV output; microgrid or distributed generation asset; islanding capability)	Energy	In development — DI-7 closed	NERC CIP (where applicable); FERC; state PUC jurisdiction; IEEE 1547 (DER interconnection)
—	DBA-ES-CntlCo	ISO, RTO, and balancing authority operational control function. FUNCTIONAL CLASS — instrument type is a functional capability instrument, not a physical facility-class design basis. Cuts across GenCo, TransCo, and DisCo.	Energy	Architecture defined	NERC functional registration (BA, RC, TOP); FERC Order 693; NERC CIP-008/014
—	DBA-ES-	Coal-fired	Energy	Active --	NERC CIP;

	GN-FC6	generation (>750 MW individual unit nameplate, or RTO/ISO must-run or transmission-essential designation)		v0.1	FERC; EPA CCR Rule (40 CFR Part 257); TSA voluntary
—	DBA-ES-GN-FC7	Oil-fired generation — distillate and residual fuel peakers (>100 MW individual unit nameplate, or RTO/ISO reliability must-run designation)	Energy	Active -- v0.1	NERC CIP; ISO/RTO RMR agreements; EPA Clean Air Act NESHAP; TSA voluntary
—	DBA-ES-GN-FC8	Renewable generation and battery energy storage — utility-scale solar PV (>200 MW cluster), onshore and offshore wind (>200 MW cluster), BESS (>100 MW / 400 MWh)	Energy	Active -- v0.1	NERC CIP; NERC CIP-013 (supply chain); FERC Order 841; BOEM (offshore); DOE EO 13920 Prohibition Order authority
—	0.9	June 2026	T. Roxey	Housekeeping corrections — stale forward-looking language updated. (1) §1.5 CntlCo paragraph	

				: language stating “no DBA-ES facility-class instrument currently addresses CntlCo infrastructure” and “The CntlCo MA extension will be developed in that instrument” corrected to reflect that DBA-ES-CntlCo v0.2 is a complete three-layer instrument with SABD metric, full DBA consequence envelope, and MA extension including the MA-1 state estimator corruption scenario.	
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				(2) DBA-MA companion paragraph : “will emerge as the facility-class instruments mature” updated to reflect that all facility-class instruments have reached maturity and DBA-ES-MA Phase B is the next registered work item (Gap 2.4). No substantive analytical changes.	
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DBA-DC is noted as the anchor instrument because it was developed before this sector-level parent document existed and was the first facility-class application of a design-basis methodology to the electric sector. Its internal methodology layer (Section 3 of DBA-DC v1.0) is the seed from which this sector-level document was developed. Where any inconsistency exists between DBA-DC’s internal methodology formulation and this sector-level document, this document governs, and the inconsistency is an open item for resolution at the next DBA-DC revision.

8.4 Development Sequence

The facility class instruments are being developed in the following sequence, driven by a combination of consequence severity, regulatory engagement maturity, and analytical complexity:

DBA-ES-HY-FC1 (large hydroelectric storage) is the next instrument to be developed following publication of this sector-level document. Large hydro presents the clearest regulatory analog to the nuclear sector DBA experience — FERC Part 12 licensing, NERC CIP-014 physical security requirements, and an established dam safety engineering community — and the most direct military action precedent in the open literature (Kakhovka, Nova Kakhovka Dam, June 2023).

DBA-ES-TX-FC2 (large EHV transmission substations) is the second facility class instrument to be developed. The Metcalf sniper attack (April 2013) and the FERC/NERC physical security response that produced CIP-014 provide the evidentiary foundation, while the long-lead-time transformer replacement problem poses the governing design-basis challenge. Development will follow completion of DBA-ES-HY-FC1.

DBA-ES-TX-FC3 (HVDC interconnects) and DBA-ES-GC-FC4 (major natural gas compressor stations for gas-electric cascade analysis) are anticipated following DBA-ES-TX-FC2. Their development sequence will be confirmed as the series matures.

8.5 The Critical Loads Subsector

The Critical Loads subsector (DBA-ES-CL) groups the facility classes in the Electric Sector series that are large electrical loads rather than generation, transmission, distribution, or control infrastructure. Where the other facility classes in this series sit on the supply, delivery, or control side of the meter, a critical load sits on the consumption side: it is an electricity consumer whose design basis is nonetheless governed by electric-sector initiating events and mitigations, and whose compromise carries consequence outward into other infrastructure sectors. The subsector exists because a large load of this kind is a different kind of object from a substation or a compressor station, and grouping it with them obscures both what governs its design basis and why it is placed in the electric sector at all.

A facility class belongs in the Critical Loads subsector when two conditions are met together. First, it is grid-material: a load large enough, or coupled to the bulk electric system closely enough, that its interconnection is a federally cognizable event and its behavior is a question in electric system engineering — the hyperscale data center threshold of one hundred megawatts of IT load is the prototype. Second, its compromise satisfies the Cross-Sector Consequence Elevation Rule at DBA-EN-SF §3.4: the consequence of its loss propagates into a dependent sector at the Tier 2 or Tier 3 level. The first condition distinguishes a critical load from an ordinary large load; the second distinguishes it from a load whose loss is contained within the electric sector. Both must hold.

The subsector is defined by load-side design-basis dominance and cross-sector elevation, not by the direction of its dependency. The hyperscale data center is a forward-only load — it consumes electricity and returns no energy-sector product — but forward-only behavior is a property of that facility class, not of the subsector. Future Critical Loads members may carry a reverse energy-sector dependency: electrolyzers and hydrogen production, large industrial process loads, dedicated AI compute beyond the hyperscale data center class, and high-density transportation-electrification hubs are candidate classes whose design basis would be electric-sector-dominated while their product or byproduct feeds another sector. The subsector definition admits these without modification.

The founding and currently sole member of the Critical Loads subsector is the hyperscale data center series, DBA-DC. Consistent with the naming convention at DBA-EN-SF §4.5, DBA-DC retains its existing designation; the Critical Loads grouping is an organizing layer within the Electric Sector series, not a renaming of the instruments it contains. The inductive lineage is preserved unchanged: DBA-DC remains the instrument from which this sector-level framework was developed, and the Critical Loads subsector, established after that development, does not stand between DBA-DC and the sector framework in that relationship.

Section 9 — Deferred Items and Future Development

9.1 Deferred Items

The following items are identified for resolution in subsequent revision passes of this document.

DI-1: Quantitative calibration of Tier 3 backup posture durations. Section 6.3 establishes the cross-sector dependency client backup posture durations at the national planning level, based on engineering judgment. These figures require validation against facility-class-specific data as the facility-class instruments are developed. Resolution criterion: updated duration figures supported by facility operator data and NERC/FERC planning analysis incorporated at next revision.

DI-2: Consistency reconciliation with DBA-DC methodology layer. As noted in Section 8.3, DBA-DC v1.0 was developed before this sector-level document and contains its own methodology layer formulation in Section 3. Any inconsistencies between DBA-DC's formulation and this document's formulation of the three required adaptations, the consequence taxonomy structure, and the gap analysis framework must be identified and resolved. Resolution criterion: DBA-DC Section 3 reviewed against this document; inconsistencies documented and resolved at next DBA-DC revision pass.

DI-3: Sector-level DBA-MA companion document. Section 5.3 anticipates a sector-level military action analysis addressing multi-facility coordinated attacks, cross-interconnection consequences, and supply chain interdiction at the national scale. That analysis is deferred pending completion of the first facility-class design bases, which will provide the facility-level analytical foundation—resolution criterion: sector-level DBA-MA companion document initiated following completion of DBA-ES-HY-FC1 and DBA-ES-TX-FC2.

DI-4: Gas sector coordination on DBA-ES-GC-FC4 scope boundary. The boundary between DBA-GAS-FC1 (natural gas transmission) and DBA-ES-GC-FC4 (compressor stations in electric generation support corridors) requires explicit coordination to ensure that the two instruments together provide complete coverage without analytical gaps or duplications. Resolution criterion: joint review of DBA-GAS-FC1 and DBA-ES-GC-FC4 scope boundaries before DBA-ES-GC-FC4 publication.

Resolved in v0.2: DI-5 (GB vs. NERC criticality criterion — resolved by broadening Tier 3 in Section 6.1 to include both the cascade pathway and the direct critical dependency pathway) and DI-6 (MA extension applicability — resolved by adding Section 1.5 with six explicit trigger criteria and the statement that facilities meeting none are DBA-complete).

DI-7: DisCo facility class gap. The Defense Science Board (“More Fight — Less Fuel,” February 2008) documented that 31 of DoD’s 34 most critical assets rely on commercial distribution grid — DisCo — infrastructure as their primary electricity source. The DBA-ES series currently has no facility-class instrument that addresses distribution-level assets serving defense-critical loads. DBA-ES-TX-FC2 addresses TransCo EHV substations; it does not reach the distribution substations and feeders that constitute the last mile to defense-critical installations. Resolution options: (a) DBA-ES-DI-FC5 designation for distribution assets serving defense-critical loads; (b) targeted annex to DBA-ES-TX-FC2 extending the TransCo substation analysis to the distribution level where military direct dependency is present. This gap is closed at v0.5. DBA-ES-DI-FC5 has been established as a dedicated facility class for distribution substations with edge generation capability and islanding function. The FC5 architecture, scope boundary, and DB-layer postulated event structure are developed in the DBA-ES FC5 and CntlCo Gap Analysis document (v0.1, June 2026). DI-7 is resolved.

9.2 Future Development

The following directions are anticipated for subsequent development of this document and the series.

The sector-level DBA-MA companion to this document — DBA-ES-MA — is the next planned instrument in the series. All ten facility-class instruments and the CntlCo functional class instrument have now reached sufficient maturity to anchor the sector-level MA analysis. DBA-ES-MA Phase B is registered as the highest-priority remaining work item in the DBA-EN Gaps and Recommendations document (Gap 2.4).

International annexes to facility-class instruments are anticipated following the pattern established in the DBA-MA nuclear series. Allied-nation electric sector facilities — particularly in NATO member states with similar grid architectures — present the same design basis methodology gaps as domestic facilities and the same urgency for DBA-MA extension. Annexes addressing specific national regulatory contexts will be developed as the series’ international engagement matures.

Lower-consequence facility class extensions — addressing DisCo substations, medium-scale GenCo hydroelectric facilities, and smaller data center tiers — are anticipated as the high-consequence facility class design bases establish the methodological precedent. The Tier 1–2 consequence boundary, established for high-consequence facility classes, will define the threshold below which lower-consequence facility class extensions are warranted.

Section 10 — References

References are listed in the order of their first appearance in the document. This references section is a series standard established at DBA-ES v0.4. Retrofits to earlier-series instruments occur at the next substantive revision of each document.

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Series Parent and Cross-Reference Documents

Roxey, T. DBA-GAS-Sector_Framework: Natural Gas Sector Design Basis Framework. Eclectic Technologies, v1.6, 2026.

Roxey, T. DBA-DC-L1: Hyperscale Data Center Design Basis Framework. Eclectic Technologies, v1.6, 2026.

Roxey, T. DBA-MA Introduction. Eclectic Technologies, v1.9, 2026.

Revision History

Version	Date	Author	Description
0.1	May 2026	T. Roxey	Initial working draft. Established sector-level parent document architecture for the Electric Sector Design Basis Series. Defined DB → DBA → DBA-MA logical progression at sector scale. Established consequence-chain organizing principle for facility class placement. Defined a facility class register with an inheritance model and released for technical review.
0.2	May 2026	T. Roxey	Added Section 1.5 establishing explicit trigger criteria for applicability of the military action extension; clarified that facilities meeting none of the six triggers are DBA-complete and the extension is not required. Broadened Section 6.1 Tier 3 consequence definition to capture direct critical dependency without cascade, incorporating the GB criticality criterion alongside the NERC cascade-based pathway. Introduced GenCo, TransCo, DisCo, and CntlCo terminology throughout. Resolved deferred items DI-5 and DI-6.
0.3	May 2026	T. Roxey	Added Defense Science Board “More Fight — Less Fuel” (February 2008) as the primary evidentiary anchor for the direct critical dependency pathway. DSB finding (31 of 34 DoD most critical assets on commercial DisCo grid) added to Section 1.4 organizing principle discussion, Section 1.5 Trigger 4, and Section 6.1 Tier 3 direct dependency pathway. Logged DI-7: DisCo facility class gap — no current DBA-ES FC instrument addresses distribution-level assets serving defense-critical loads.
0.4	May 2026	T. Roxey	Added Section 10 — References. Established a formal references section as a series standard. Retrofit to earlier series instruments to occur at each document’s next substantive revision—post-Grammarly clean build incorporating all v0.3 corrections.
0.5	June	T.	Added DBA-ES-DI-FC5 (distribution substation with edge

	2026	Roxey	generation) and DBA-ES-CntlCo (functional class) to the facility class register, closing DI-7. Added Section 3.4 establishing the dual Design Basis architecture (sector-specific physical DB and AI Governance DB) as a series requirement; renumbered former Section 3.4 to 3.5. Added RT_HVLLC (Recovery Timeline for High-Value Long-Lead Components) definition to Section 6.2 as a series standard consequence-duration metric, with attribution to GenMix Optimization Whitepaper—updated Section 1.5 CntlCo applicability note to reference the CntlCo functional class register entry. Formally closed DI-7 in the deferred items register.
0.6	June 2026	T. Roxey	Adopted series-wide facility class naming convention incorporating type codes. DBA-HY-FC1 renamed to DBA-ES-HY-FC1 (HY = hydroelectric). DBA-ES-FC2 renamed to DBA-ES-TX-FC2 (TX = transmission). DBA-ES-FC3 renamed to DBA-ES-TX-FC3. DBA-ES-FC4 renamed to DBA-ES-GC-FC4 (GC = gas-electric cascade). DBA-ES-FC5 renamed to DBA-ES-DI-FC5 (DI = distribution). DBA-DC series unchanged. DBA-ES-CntlCo unchanged. All register entries, prose references, deferred items, and development sequence text were updated throughout. DBA-ES-HY-FC1 and DBA-ES-TX-FC2 instrument files were renamed concurrently.
0.7	June 2026	T. Roxey	Housekeeping: References section corrected. DBA-GAS-Sector_Framework updated from v1.2 to v1.4. DBA-DC-L1 updated from v1.0 to v1.6. DBA-MA Introduction updated from v1.7 to v1.9. Body text §1.4 facility class definition updated: DBA-DC version reference corrected from v1.0 to v1.6. Historical references to DBA-DC v1.0 in §8.3 and DI-2 (describing the development sequence in which DBA-DC preceded this document) are preserved as accurate historical record. Footer version string corrected from v0.4 to v0.7.
0.8	June 2026	T. Roxey	GN sub-series established. Three new facility class instruments added to register: DBA-ES-GN-FC6 (coal-fired generation, Active v0.1), DBA-ES-GN-FC7 (oil-fired generation and peakers, Active v0.1), DBA-ES-GN-FC8 (renewable generation and BESS, Active v0.1). GN type code introduced for the GenCo generation sub-series. OI-2 (DBA-EN-SF) resolved: coal, oil-fired, and renewable GenCo FCs now at first issue. “Bright Line” corrected to “Bright Line” in §3.5 body text. No other substantive content changes.
0.9	June 2026	T. Roxey	Version aligned to 0.9 during the DBA-EN-SF parent-registration pass (recorded at DBA-EN-SF v0.9 §1.4). Registration-consistency version bump; no facility-class register or analytical content changes recorded at this revision.

0.10	June 2026	T. Roxey	Critical Loads subsector (DBA-ES-CL) established. §8.3 facility class register: Subsector column added, designating each facility class's subsector grouping (DBA-DC marked Critical Loads (DBA-ES-CL)). §8.5 added (The Critical Loads Subsector): defines the load-side subsector and its two-part membership gate (grid-material; Cross-Sector Consequence Elevation Rule at DBA-EN-SF §3.4), identifies candidate future members, and preserves the DBA-DC inductive-seed lineage and existing designation per DBA-EN-SF §4.5. DBA-DC registered as founding and sole current member. No instrument renaming.
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